Microsoft Elevate

Students Social Media Addiction Analysis Using Power BI

PRESENTED BY

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OUTLINE:

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result (Output Image)**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT:

Social media platforms such as Instagram, YouTube, and Facebook are widely used by students. Excessive use of social media can lead to addiction, reduced academic performance, sleep issues, and mental health problems.

Analyzing large datasets of student behavior manually is difficult. Therefore, there is a need for a data analytics solution to identify patterns of social media addiction and its impact on students' academic performance, sleep patterns, and mental health.

PROPOSED SOLUTION:

The proposed system focuses on analyzing students' social media usage patterns using interactive data visualization techniques in Microsoft Power BI Desktop.

The solution involves collecting a dataset that contains information about students' daily social media usage, addiction scores, sleep patterns, academic performance, and mental health indicators. The dataset is processed and cleaned to ensure accuracy and consistency.

After data preprocessing, several analytical measures are created using DAX to calculate key metrics such as total students, average daily usage hours, average academic scores, and the number of heavy social media users.

The processed data is then visualized using Power BI dashboards, which include charts and KPI indicators such as platform usage analysis, addiction level distribution, social media impact on academic performance, and the relationship between sleep hours and mental health.

These interactive visualizations help in identifying patterns and trends in students' behavior and provide meaningful insights into how excessive social media usage affects students' academic performance, mental health, and lifestyle habits.

SYSTEM APPROACH:

Technologies Used :

- Microsoft Power BI Desktop
- Excel / CSV Dataset
- Power Query – Data Cleaning
- DAX – Data Analysis Expressions
- Data Visualization

System Workflow

- Import dataset into Power BI
- Clean and transform data using Power Query
- Create calculated measures using DAX
- Build visualizations and dashboards
- Generate insights from data

ALGORITHM & DEPLOYMENT:

1. Data Collection – Social media usage dataset of students.
2. Data Cleaning – Removed duplicates and corrected data types.
3. Data Transformation – Prepared fields for analysis.
4. Data Analysis – Created DAX measures such as Total Students and Heavy Users.

Examples :

Total Students = COUNT('Students Social Media Addiction'[Age])

Heavy Users = CALCULATE(COUNT('Students Social Media Addiction'[Student_ID]),'Students Social Media Addiction'[Addicted_Score] > 7)

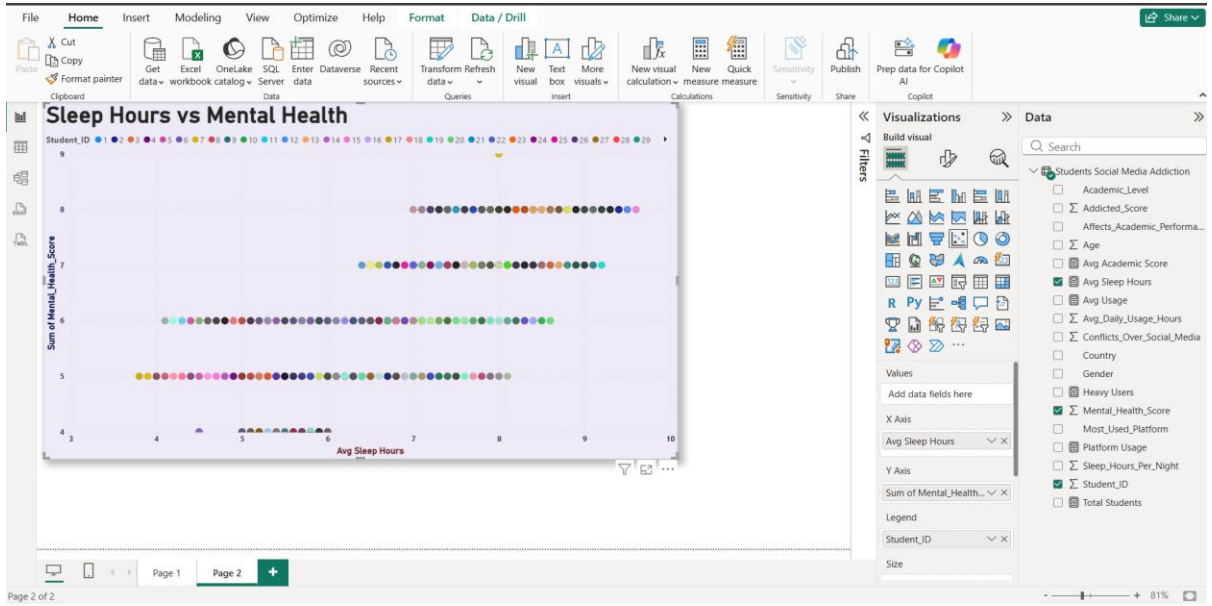
Platform Usage = COUNT('Students Social Media Addiction'[Most_Used_Platform])

5. Data Visualization – Built interactive dashboards using Power BI visuals.
6. Insight Generation – Identified patterns in usage behavior and academic performance.

RESULT:

The Power BI dashboard provides the following insights:

- Total number of students analyzed
- Average daily social media usage hours
- Distribution of addiction levels among students
- Impact of social media usage on academic performance
- Relationship between sleep hours and mental health



CONCLUSION:

- ✓This project successfully demonstrates how data analytics and visualization tools can be used to analyze social media addiction among students.
- ✓Using Power BI dashboards, the study highlights key insights such as the average daily social media usage among students, the distribution of addiction levels, and the identification of heavy social media users. The analysis also reveals relationships between social media usage, academic performance, sleep duration, and mental health indicators.
- ✓The interactive dashboard enables easy exploration of the dataset through filters and visualizations, helping users understand patterns and trends effectively. The results indicate that excessive social media usage may negatively impact students' academic performance and well-being.
- ✓Overall, this project demonstrates the importance of using data analytics to better understand behavioral patterns and provides a useful tool for researchers, educators, and policymakers to evaluate the impact of social media on students

FUTURE SCOPE:

- Although this project provides useful insights into students' social media addiction, there are several opportunities for future improvements and extensions.
- In the future, the system can be enhanced by integrating real-time data from social media platforms through APIs, allowing continuous monitoring of student behavior. Advanced machine learning algorithms can also be implemented to predict addiction levels or identify students who may be at risk of excessive social media usage.
- Sentiment analysis techniques can be applied to analyze students' posts and comments to understand emotional and psychological patterns. Additionally, the analysis can be expanded to include more demographic variables such as geographic regions, educational institutions, and age groups to provide a broader understanding of social media behavior.
- The system can also be developed into a web-based dashboard that educators and researchers can use for monitoring and analysis. These improvements would make the solution more powerful and useful for addressing the challenges related to social media addiction among students.

REFERENCES:

- Kaggle Dataset – Social Media Addiction Dataset
- Microsoft Power BI Documentation
- NDAP Open Data Portal
- Research Articles on Social Media Addiction
- Data Analytics Learning Resources

GitHub Link: Link

Thank You